KITABU QUEST

Grade 5

SCIENCE AND TECHNOLOGY

- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

GRADE 5

KEN CBC aligned draft

Title Page

KITABU QUEST Grade 5 Science and Technology

Kenya CBC content draft for review

Mascot: chameleon

This content-first draft was regenerated from Kitabu curriculum data using a topic-by-topic book plan. It is designed to help learners read, practise, discuss, create, and revise with confidence.

Cover artwork is intentionally deferred until all content has passed review and is marked ready-for-cover.

How To Use This Book

This book helps Grade 5 learners study Science and Technology through clear lessons, worked examples, activities, practice, and revision.

Use the inquiry questions to think first. Read the explanation. Try the guided example. Complete the activity. Answer the practice questions. Correct your work and ask for support where needed.

Observe, ask questions, test safely, record evidence, and explain findings.

Every unit has a local example, a misconception to avoid, success criteria, and a check for understanding.

Learning Skills In This Book

Each chapter builds knowledge, skills, values, creativity, communication, collaboration, critical thinking, digital literacy, and self-confidence.

Look for these page types: Learn, Example, Activity, Practice, Reflection, Home Link, and Review.

How to study well: read the goal first, underline new words, try the example before reading the answer, discuss with a partner, and correct your work after feedback. Do not skip the reflection questions. They help you notice what you understand and what needs more practice.

Table Of Contents

Use this table of contents to plan your study. Start with the first chapter, but return to earlier pages whenever you need revision.

After each chapter, complete the chapter review before moving forward. At the end of the book, use the glossary, answer notes, and final project to revise the whole subject.

Chapter: Living Things and their Environment

In this chapter you will study Living Things and their Environment.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:
- 1.1 Classification of Plants
- 1.2 The Human Breathing System

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Classification of Plants: classify plants into flowering and non-flowering: Lesson Opener

Learning area: Classification of Plants: classify plants into flowering and non-flowering

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Classification of Plants: classify plants into flowering and non-flowering.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on Classification of Plants: classify plants into flowering and non-flowering.

By the end of this lesson sequence, you should be able to:
- classify plants into flowering and non-flowering
- describe functions of parts of a flower
- outline the importance of flowers in nature

Inquiry questions:
- How are plants classified?

Before reading, write one thing you already know and one question you want this lesson to answer.

Classification of Plants: classify plants into flowering and non-flowering: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Classification of Plants: classify plants into flowering and non-flowering.

Important idea: Classification of Plants: classify plants into flowering and non-flowering.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Classification of Plants: classify plants into flowering and non-flowering
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- Classification of Plants: classify plants into flowering and non-flowering
- Classification of Plants
- Classification
- Plants
- classify
- flowering
- non-flowering

Explain the idea in your own words before attempting the activities.

Classification of Plants: classify plants into flowering and non-flowering: Worked Example

Investigation model for Classification of Plants: classify plants into flowering and non-flowering: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:
1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Classification of Plants: classify plants into flowering and non-flowering: Activity

Activity for Classification of Plants: classify plants into flowering and non-flowering: Plan a safe observation or investigation.

Inquiry question: How are plants classified?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Classification of Plants: classify plants into flowering and non-flowering: Activity And Practice

Practice for Classification of Plants: classify plants into flowering and non-flowering:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Classify six plants into flowering and non-flowering groups using visible evidence. Example table: hibiscus, maize, and bean are flowering plants because they produce flowers or seeds; fern and moss are non-flowering plants because they reproduce using spores; pine is non-flowering but produces cones. Answer guide: classification must name the plant and the observed feature. Target outcome: classify plants into flowering and non-flowering. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Draw a simple flower and label petals, sepals, stamen, pistil, ovary, and pollen. Write one function for each part. Answer guide: petals attract pollinators, sepals protect the bud, stamen produces pollen, pistil receives pollen, and ovary develops seeds after fertilisation. Target outcome: describe functions of parts of a flower. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Explain three roles of flowers in nature using examples: they support reproduction in flowering plants, attract pollinators such as bees or butterflies, and lead to fruits or seeds that feed people and animals. Answer guide: each role must connect the flower part or process to a benefit in the ecosystem. Target outcome: outline the importance of flowers in nature. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Classification of Plants: classify plants into flowering and non-flowering: Outcome Check

Outcome: classify plants into flowering and non-flowering

Show mastery with topic evidence.

Task: Classify six plants into flowering and non-flowering groups using visible evidence. Example table: hibiscus, maize, and bean are flowering plants because they produce flowers or seeds; fern and moss are non-flowering plants because they reproduce using spores; pine is non-flowering but produces cones. Answer guide: classification must name the plant and the observed feature.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Classification of Plants: classify plants into flowering and non-flowering, uses correct science vocabulary, and connects evidence to explanation.

Classification of Plants: classify plants into flowering and non-flowering: Outcome Check

Outcome: describe functions of parts of a flower

Show mastery with topic evidence.

Task: Draw a simple flower and label petals, sepals, stamen, pistil, ovary, and pollen. Write one function for each part. Answer guide: petals attract pollinators, sepals protect the bud, stamen produces pollen, pistil receives pollen, and ovary develops seeds after fertilisation.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Classification of Plants: classify plants into flowering and non-flowering, uses correct science vocabulary, and connects evidence to explanation.

Classification of Plants: classify plants into flowering and non-flowering: Outcome Check

Outcome: outline the importance of flowers in nature

Show mastery with topic evidence.

Task: Explain three roles of flowers in nature using examples: they support reproduction in flowering plants, attract pollinators such as bees or butterflies, and lead to fruits or seeds that feed people and animals. Answer guide: each role must connect the flower part or process to a benefit in the ecosystem.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Classification of Plants: classify plants into flowering and non-flowering, uses correct science vocabulary, and connects evidence to explanation.

Classification of Plants: flowers in nature: Lesson Opener

Learning area: Classification of Plants: flowers in nature

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Classification of Plants: flowers in nature.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on Classification of Plants: flowers in nature.

By the end of this lesson sequence, you should be able to:
- flowers in nature

Inquiry questions:
- How are plants classified?

Before reading, write one thing you already know and one question you want this lesson to answer.

Classification of Plants: flowers in nature: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Classification of Plants: flowers in nature.

Important idea: Classification of Plants: flowers in nature.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Classification of Plants: flowers in nature
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- Classification of Plants: flowers in nature
- Classification of Plants
- Classification
- Plants
- flowers
- nature

Explain the idea in your own words before attempting the activities.

Classification of Plants: flowers in nature: Worked Example

Investigation model for Classification of Plants: flowers in nature: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:
1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Classification of Plants: flowers in nature: Activity

Activity for Classification of Plants: flowers in nature: Plan a safe observation or investigation.

Inquiry question: How are plants classified?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Classification of Plants: flowers in nature: Activity And Practice

Practice for Classification of Plants: flowers in nature:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Explain three roles of flowers in nature using examples: they support reproduction in flowering plants, attract pollinators such as bees or butterflies, and lead to fruits or seeds that feed people and animals. Answer guide: each role must connect the flower part or process to a benefit in the ecosystem. Target outcome: flowers in nature. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Classification of Plants: flowers in nature: Outcome Check

Outcome: flowers in nature

Show mastery with topic evidence.

Task: Explain three roles of flowers in nature using examples: they support reproduction in flowering plants, attract pollinators such as bees or butterflies, and lead to fruits or seeds that feed people and animals. Answer guide: each role must connect the flower part or process to a benefit in the ecosystem.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Classification of Plants: flowers in nature, uses correct science vocabulary, and connects evidence to explanation.

The Human Breathing System: main parts of the human breathing system: Lesson Opener

Learning area: The Human Breathing System: main parts of the human breathing system

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to The Human Breathing System: main parts of the human breathing system.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on The Human Breathing System: main parts of the human breathing system.

By the end of this lesson sequence, you should be able to:
- identify the main parts of the human breathing system
- describe the functions of parts of the human breathing system
- coutline the symptoms and prevention measures for common conditions and diseases of the breathing system

Inquiry questions:
- What makes up the human breathing system?
- What enhances a healthy breathing system?

Before reading, write one thing you already know and one question you want this lesson to answer.

The Human Breathing System: main parts of the human breathing system: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to The Human Breathing System: main parts of the human breathing system.

Important idea: The Human Breathing System: main parts of the human breathing system.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in The Human Breathing System: main parts of the human breathing system
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- The Human Breathing System: main parts of the human breathing system
- The Human Breathing System
- Human
- Breathing
- System
- parts

Explain the idea in your own words before attempting the activities.

The Human Breathing System: main parts of the human breathing system: Worked Example

Investigation model for The Human Breathing System: main parts of the human breathing system: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:
1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

The Human Breathing System: main parts of the human breathing system: Activity

Activity for The Human Breathing System: main parts of the human breathing system: Plan a safe observation or investigation.

Inquiry question: What makes up the human breathing system?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

The Human Breathing System: main parts of the human breathing system: Activity And Practice

Practice for The Human Breathing System: main parts of the human breathing system:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: identify the main parts of the human breathing system. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: describe the functions of parts of the human breathing system. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: coutline the symptoms and prevention measures for common conditions and diseases of the breathing system. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

The Human Breathing System: main parts of the human breathing system: Outcome Check

Outcome: identify the main parts of the human breathing system

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in The Human Breathing System: main parts of the human breathing system, uses correct science vocabulary, and connects evidence to explanation.

The Human Breathing System: main parts of the human breathing system: Outcome Check

Outcome: describe the functions of parts of the human breathing system

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in The Human Breathing System: main parts of the human breathing system, uses correct science vocabulary, and connects evidence to explanation.

The Human Breathing System: main parts of the human breathing system: Outcome Check

Outcome: coutline the symptoms and prevention measures for common conditions and diseases of the breathing system

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in The Human Breathing System: main parts of the human breathing system, uses correct science vocabulary, and connects evidence to explanation.

The Human Breathing System: need for maintaining a healthy breathing system.: Lesson Opener

Learning area: The Human Breathing System: need for maintaining a healthy breathing system.

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to The Human Breathing System: need for maintaining a healthy breathing system.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on The Human Breathing System: need for maintaining a healthy breathing system.

By the end of this lesson sequence, you should be able to:
- appreciate the need for maintaining a healthy breathing system.

Inquiry questions:
- What enhances a healthy breathing system?

Before reading, write one thing you already know and one question you want this lesson to answer.

The Human Breathing System: need for maintaining a healthy breathing system.: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to The Human Breathing System: need for maintaining a healthy breathing system.

Important idea: The Human Breathing System: need for maintaining a healthy breathing system.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in The Human Breathing System: need for maintaining a healthy breathing system.
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- The Human Breathing System: need for maintaining a healthy breathing system.
- The Human Breathing System
- Human
- Breathing
- System
- maintaining
- healthy
- system.

Explain the idea in your own words before attempting the activities.

The Human Breathing System: need for maintaining a healthy breathing system.: Worked Example

Investigation model for The Human Breathing System: need for maintaining a healthy breathing system.: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:
1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

The Human Breathing System: need for maintaining a healthy breathing system.: Activity

Activity for The Human Breathing System: need for maintaining a healthy breathing system.: Plan a safe observation or investigation.

Inquiry question: What enhances a healthy breathing system?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

The Human Breathing System: need for maintaining a healthy breathing system.: Activity And Practice

Practice for The Human Breathing System: need for maintaining a healthy breathing system.:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: appreciate the need for maintaining a healthy breathing system. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

The Human Breathing System: need for maintaining a healthy breathing system.: Outcome Check

Outcome: appreciate the need for maintaining a healthy breathing system.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in The Human Breathing System: need for maintaining a healthy breathing system., uses correct science vocabulary, and connects evidence to explanation.

Living Things and their Environment: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving,

In this chapter you will study Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:
- 2.0 Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous: Lesson Opener

Learning area: Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous.

By the end of this lesson sequence, you should be able to:
- classify mixtures as homogeneous or heterogeneous
- apply appropriate methods to separate heterogeneous mixtures
- outline the applications of separating mixtures in day-to-day life

Inquiry questions:
- What are the applications of mixture separation in day-to-day life?
- What are the dangers of water pollution?

Before reading, write one thing you already know and one question you want this lesson to answer.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous.

Important idea: Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous.

Materials can be described by observable properties such as state, texture, solubility, magnetism, colour, and reaction. Choose a separation or testing method because it matches a property, not because the method sounds familiar.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous
- Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving
- Matter
- Mixtures
- Meaning
- Types
- Winnowing
- picking

Explain the idea in your own words before attempting the activities.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous: Worked Example

Investigation model for Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous: Compare material properties.

Question: Which property helps us identify or separate the materials?

Example: A mixture of sand and salt can be separated because salt dissolves in water but sand does not. The sand can be filtered out, and salt can be recovered from the solution by evaporation.

Procedure:
1. List the materials and their visible properties.
2. Predict which property will be useful: size, solubility, magnetism, boiling point, texture, or colour change.
3. Choose a safe method that matches the property.
4. Record what happens before and after each step.
5. Explain why the method worked instead of only naming the method.

Conclusion frame: The useful property was _____, so the best method was _____.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous: Activity

Activity for Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous: Plan a safe observation or investigation.

Inquiry question: What are the applications of mixture separation in day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous: Activity And Practice

Practice for Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: classify mixtures as homogeneous or heterogeneous. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: apply appropriate methods to separate heterogeneous mixtures. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: outline the applications of separating mixtures in day-to-day life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous: Outcome Check

Outcome: classify mixtures as homogeneous or heterogeneous

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous: Outcome Check

Outcome: apply appropriate methods to separate heterogeneous mixtures

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous: Outcome Check

Outcome: outline the applications of separating mixtures in day-to-day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: classify mixtures as homogeneous, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating: Lesson Opener

Learning area: Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating.

By the end of this lesson sequence, you should be able to:
- appreciate different methods of separating mixtures in day-to-day life.
- identify water pollutants in the water sources
- outline the effects of water pollution in day-to-day life

Inquiry questions:
- What are the dangers of water pollution?

Before reading, write one thing you already know and one question you want this lesson to answer.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating.

Important idea: Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating.

Materials can be described by observable properties such as state, texture, solubility, magnetism, colour, and reaction. Choose a separation or testing method because it matches a property, not because the method sounds familiar.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating
- Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving
- Matter
- Mixtures
- Meaning
- Types
- Winnowing
- picking

Explain the idea in your own words before attempting the activities.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating: Worked Example

Investigation model for Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating: Compare material properties.

Question: Which property helps us identify or separate the materials?

Example: A mixture of sand and salt can be separated because salt dissolves in water but sand does not. The sand can be filtered out, and salt can be recovered from the solution by evaporation.

Procedure:
1. List the materials and their visible properties.
2. Predict which property will be useful: size, solubility, magnetism, boiling point, texture, or colour change.
3. Choose a safe method that matches the property.
4. Record what happens before and after each step.
5. Explain why the method worked instead of only naming the method.

Conclusion frame: The useful property was _____, so the best method was _____.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating: Activity

Activity for Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating: Plan a safe observation or investigation.

Inquiry question: What are the dangers of water pollution?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating: Activity And Practice

Practice for Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: appreciate different methods of separating mixtures in day-to-day life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: identify water pollutants in the water sources. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: outline the effects of water pollution in day-to-day life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating: Outcome Check

Outcome: appreciate different methods of separating mixtures in day-to-day life.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating: Outcome Check

Outcome: identify water pollutants in the water sources

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
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Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating: Outcome Check

Outcome: outline the effects of water pollution in day-to-day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: different methods of separating, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water: Lesson Opener

Learning area: Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water.

By the end of this lesson sequence, you should be able to:
- identify methods of reducing water pollution in the water sources
- apply appropriate methods of water treatment
- advocate for safe water sources in the locality.

Inquiry questions:
- What are the applications of mixture separation in day-to-day life?
- What are the dangers of water pollution?

Before reading, write one thing you already know and one question you want this lesson to answer.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water.

Important idea: Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water.

Materials can be described by observable properties such as state, texture, solubility, magnetism, colour, and reaction. Choose a separation or testing method because it matches a property, not because the method sounds familiar.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water
- Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving
- Matter
- Mixtures
- Meaning
- Types
- Winnowing
- picking

Explain the idea in your own words before attempting the activities.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water: Worked Example

Investigation model for Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water: Compare material properties.

Question: Which property helps us identify or separate the materials?

Example: A mixture of sand and salt can be separated because salt dissolves in water but sand does not. The sand can be filtered out, and salt can be recovered from the solution by evaporation.

Procedure:
1. List the materials and their visible properties.
2. Predict which property will be useful: size, solubility, magnetism, boiling point, texture, or colour change.
3. Choose a safe method that matches the property.
4. Record what happens before and after each step.
5. Explain why the method worked instead of only naming the method.

Conclusion frame: The useful property was _____, so the best method was _____.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water: Activity

Activity for Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water: Plan a safe observation or investigation.

Inquiry question: What are the applications of mixture separation in day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water: Activity And Practice

Practice for Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: identify methods of reducing water pollution in the water sources. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: apply appropriate methods of water treatment. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: advocate for safe water sources in the locality. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water: Outcome Check

Outcome: identify methods of reducing water pollution in the water sources

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water: Outcome Check

Outcome: apply appropriate methods of water treatment

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water: Outcome Check

Outcome: advocate for safe water sources in the locality.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving,: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Force and Energy

In this chapter you will study Force and Energy.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:
- 3.1 Floating
- 3.2 Sound Energy
- 3.3 Heat transfer

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Floating: Lesson Opener

Learning area: Floating

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Floating.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on Floating.

By the end of this lesson sequence, you should be able to:
- effects of flooding and the use of floaters as life savers
- Project: Learners to make lifesaving floaters from locally available materials.

Inquiry questions:
- Why do some materials float and others sink?
- How are floaters useful in day-to-day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Floating: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Floating.

Important idea: Floating.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Floating
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- Floating

Explain the idea in your own words before attempting the activities.

Floating: Worked Example

Investigation model for Floating: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:
1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Floating: Activity

Activity for Floating: Plan a safe observation or investigation.

Inquiry question: Why do some materials float and others sink?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Floating: Activity And Practice

Practice for Floating:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: effects of flooding and the use of floaters as life savers. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: Project: Learners to make lifesaving floaters from locally available materials. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Floating: Outcome Check

Outcome: effects of flooding and the use of floaters as life savers

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Floating, uses correct science vocabulary, and connects evidence to explanation.

Floating: Outcome Check

Outcome: Project: Learners to make lifesaving floaters from locally available materials.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Floating, uses correct science vocabulary, and connects evidence to explanation.

Sound Energy: sources of sound in nature: Lesson Opener

Learning area: Sound Energy: sources of sound in nature

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Sound Energy: sources of sound in nature.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on Sound Energy: sources of sound in nature.

By the end of this lesson sequence, you should be able to:
- identify sources of sound in nature
- demonstrate the movement of sound in nature
- describe effects of loud sound in day-to-day life

Inquiry questions:
- How is sound produced?
- What are the effects of loud sound?

Before reading, write one thing you already know and one question you want this lesson to answer.

Sound Energy: sources of sound in nature: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Sound Energy: sources of sound in nature.

Important idea: Sound Energy: sources of sound in nature.

Physical science looks for patterns in energy, forces, and materials. Keep one condition changing at a time, observe the effect, and use measured evidence where possible. A correct explanation names the cause, the change observed, and the reason the change happened.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Sound Energy: sources of sound in nature
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- Sound Energy: sources of sound in nature
- Sound Energy
- Sound
- Energy
- sources
- nature

Explain the idea in your own words before attempting the activities.

Sound Energy: sources of sound in nature: Worked Example

Investigation model for Sound Energy: sources of sound in nature: Test one energy or circuit condition safely.

Question: What changes when one condition is changed?

Example: In a simple circuit, a bulb lights only when there is a complete path from one battery terminal, through the bulb, and back to the other terminal.

Procedure:
1. Name the variable you will change, such as number of cells, wire length, material, distance, or surface.
2. Name what you will observe or measure.
3. Keep the other conditions the same.
4. Record results in a table with at least three trials or comparisons.
5. Write a conclusion that mentions the pattern in the results.

Safety rule: Use only approved school materials and ask before connecting any electrical device.

Sound Energy: sources of sound in nature: Activity

Activity for Sound Energy: sources of sound in nature: Plan a safe observation or investigation.

Inquiry question: How is sound produced?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Sound Energy: sources of sound in nature: Activity And Practice

Practice for Sound Energy: sources of sound in nature:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: identify sources of sound in nature. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: demonstrate the movement of sound in nature. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: describe effects of loud sound in day-to-day life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Sound Energy: sources of sound in nature: Outcome Check

Outcome: identify sources of sound in nature

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Sound Energy: sources of sound in nature, uses correct science vocabulary, and connects evidence to explanation.

Sound Energy: sources of sound in nature: Outcome Check

Outcome: demonstrate the movement of sound in nature

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Sound Energy: sources of sound in nature, uses correct science vocabulary, and connects evidence to explanation.

Sound Energy: sources of sound in nature: Outcome Check

Outcome: describe effects of loud sound in day-to-day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Sound Energy: sources of sound in nature, uses correct science vocabulary, and connects evidence to explanation.

Sound Energy: role of sound in day-to-day life.: Lesson Opener

Learning area: Sound Energy: role of sound in day-to-day life.

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Sound Energy: role of sound in day-to-day life.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on Sound Energy: role of sound in day-to-day life.

By the end of this lesson sequence, you should be able to:
- appreciate the role of sound in day-to-day life.

Inquiry questions:
- What are the effects of loud sound?

Before reading, write one thing you already know and one question you want this lesson to answer.

Sound Energy: role of sound in day-to-day life.: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Sound Energy: role of sound in day-to-day life.

Important idea: Sound Energy: role of sound in day-to-day life.

Physical science looks for patterns in energy, forces, and materials. Keep one condition changing at a time, observe the effect, and use measured evidence where possible. A correct explanation names the cause, the change observed, and the reason the change happened.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Sound Energy: role of sound in day-to-day life.
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- Sound Energy: role of sound in day-to-day life.
- Sound Energy
- Sound
- Energy
- day-to-day
- life.

Explain the idea in your own words before attempting the activities.

Sound Energy: role of sound in day-to-day life.: Worked Example

Investigation model for Sound Energy: role of sound in day-to-day life.: Test one energy or circuit condition safely.

Question: What changes when one condition is changed?

Example: In a simple circuit, a bulb lights only when there is a complete path from one battery terminal, through the bulb, and back to the other terminal.

Procedure:
1. Name the variable you will change, such as number of cells, wire length, material, distance, or surface.
2. Name what you will observe or measure.
3. Keep the other conditions the same.
4. Record results in a table with at least three trials or comparisons.
5. Write a conclusion that mentions the pattern in the results.

Safety rule: Use only approved school materials and ask before connecting any electrical device.

Sound Energy: role of sound in day-to-day life.: Activity

Activity for Sound Energy: role of sound in day-to-day life.: Plan a safe observation or investigation.

Inquiry question: What are the effects of loud sound?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Sound Energy: role of sound in day-to-day life.: Activity And Practice

Practice for Sound Energy: role of sound in day-to-day life.:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: appreciate the role of sound in day-to-day life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Sound Energy: role of sound in day-to-day life.: Outcome Check

Outcome: appreciate the role of sound in day-to-day life.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Sound Energy: role of sound in day-to-day life., uses correct science vocabulary, and connects evidence to explanation.

Heat transfer: modes of heat transfer in nature: Lesson Opener

Learning area: Heat transfer: modes of heat transfer in nature

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Heat transfer: modes of heat transfer in nature.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on Heat transfer: modes of heat transfer in nature.

By the end of this lesson sequence, you should be able to:
- demonstrate the modes of heat transfer in nature
- classify conductors of heat into good or poor conductors
- explain the uses of heat transfer in day-to-day life

Inquiry questions:
- How is heat transferred through materials in nature?

Before reading, write one thing you already know and one question you want this lesson to answer.

Heat transfer: modes of heat transfer in nature: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Heat transfer: modes of heat transfer in nature.

Important idea: Heat transfer: modes of heat transfer in nature.

Physical science looks for patterns in energy, forces, and materials. Keep one condition changing at a time, observe the effect, and use measured evidence where possible. A correct explanation names the cause, the change observed, and the reason the change happened.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Heat transfer: modes of heat transfer in nature
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- Heat transfer: modes of heat transfer in nature
- Heat transfer
- transfer
- modes
- nature

Explain the idea in your own words before attempting the activities.

Heat transfer: modes of heat transfer in nature: Worked Example

Investigation model for Heat transfer: modes of heat transfer in nature: Test one energy or circuit condition safely.

Question: What changes when one condition is changed?

Example: In a simple circuit, a bulb lights only when there is a complete path from one battery terminal, through the bulb, and back to the other terminal.

Procedure:
1. Name the variable you will change, such as number of cells, wire length, material, distance, or surface.
2. Name what you will observe or measure.
3. Keep the other conditions the same.
4. Record results in a table with at least three trials or comparisons.
5. Write a conclusion that mentions the pattern in the results.

Safety rule: Use only approved school materials and ask before connecting any electrical device.

Heat transfer: modes of heat transfer in nature: Activity

Activity for Heat transfer: modes of heat transfer in nature: Plan a safe observation or investigation.

Inquiry question: How is heat transferred through materials in nature?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Heat transfer: modes of heat transfer in nature: Activity And Practice

Practice for Heat transfer: modes of heat transfer in nature:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: demonstrate the modes of heat transfer in nature. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: classify conductors of heat into good or poor conductors. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: explain the uses of heat transfer in day-to-day life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Heat transfer: modes of heat transfer in nature: Outcome Check

Outcome: demonstrate the modes of heat transfer in nature

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Heat transfer: modes of heat transfer in nature, uses correct science vocabulary, and connects evidence to explanation.

Heat transfer: modes of heat transfer in nature: Outcome Check

Outcome: classify conductors of heat into good or poor conductors

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Heat transfer: modes of heat transfer in nature, uses correct science vocabulary, and connects evidence to explanation.

Heat transfer: modes of heat transfer in nature: Outcome Check

Outcome: explain the uses of heat transfer in day-to-day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Heat transfer: modes of heat transfer in nature, uses correct science vocabulary, and connects evidence to explanation.

Heat transfer: acknowledge safety precautions when handling heat.: Lesson Opener

Learning area: Heat transfer: acknowledge safety precautions when handling heat.

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Heat transfer: acknowledge safety precautions when handling heat.

Curriculum link: This lesson follows the Grade 5 curriculum sequence and focuses on Heat transfer: acknowledge safety precautions when handling heat.

By the end of this lesson sequence, you should be able to:
- acknowledge safety precautions when handling heat.

Inquiry questions:
- How is heat transferred through materials in nature?

Before reading, write one thing you already know and one question you want this lesson to answer.

Heat transfer: acknowledge safety precautions when handling heat.: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Heat transfer: acknowledge safety precautions when handling heat.

Important idea: Heat transfer: acknowledge safety precautions when handling heat.

Physical science looks for patterns in energy, forces, and materials. Keep one condition changing at a time, observe the effect, and use measured evidence where possible. A correct explanation names the cause, the change observed, and the reason the change happened.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 5, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Heat transfer: acknowledge safety precautions when handling heat.
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:
- Heat transfer: acknowledge safety precautions when handling heat.
- Heat transfer
- transfer
- acknowledge
- safety
- precautions
- handling
- heat.

Explain the idea in your own words before attempting the activities.

Heat transfer: acknowledge safety precautions when handling heat.: Worked Example

Investigation model for Heat transfer: acknowledge safety precautions when handling heat.: Test one energy or circuit condition safely.

Question: What changes when one condition is changed?

Example: In a simple circuit, a bulb lights only when there is a complete path from one battery terminal, through the bulb, and back to the other terminal.

Procedure:
1. Name the variable you will change, such as number of cells, wire length, material, distance, or surface.
2. Name what you will observe or measure.
3. Keep the other conditions the same.
4. Record results in a table with at least three trials or comparisons.
5. Write a conclusion that mentions the pattern in the results.

Safety rule: Use only approved school materials and ask before connecting any electrical device.

Heat transfer: acknowledge safety precautions when handling heat.: Activity

Activity for Heat transfer: acknowledge safety precautions when handling heat.: Plan a safe observation or investigation.

Inquiry question: How is heat transferred through materials in nature?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:
- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Heat transfer: acknowledge safety precautions when handling heat.: Activity And Practice

Practice for Heat transfer: acknowledge safety precautions when handling heat.:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: acknowledge safety precautions when handling heat. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:
- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Heat transfer: acknowledge safety precautions when handling heat.: Outcome Check

Outcome: acknowledge safety precautions when handling heat.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:
1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Heat transfer: acknowledge safety precautions when handling heat., uses correct science vocabulary, and connects evidence to explanation.

Force and Energy: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Classification of Plants: classify plants into flowering and non-flowering: Review Clinic 1

Use this review clinic to strengthen a real unit from the book.

Practice context: home or school example connected to Classification of Plants: classify plants into flowering and non-flowering.

1. Re-read the lesson on Classification of Plants: classify plants into flowering and non-flowering and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: observation is specific.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

The Human Breathing System: need for maintaining a healthy breathing system.: Review Clinic 2

Use this review clinic to strengthen a real unit from the book.

Practice context: county or community example connected to The Human Breathing System: need for maintaining a healthy breathing system.

1. Re-read the lesson on The Human Breathing System: need for maintaining a healthy breathing system. and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: safety is considered.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water: Review Clinic 3

Use this review clinic to strengthen a real unit from the book.

Practice context: partner discussion about Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water.

1. Re-read the lesson on Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: methods of reducing water and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: conclusion uses evidence.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Floating: Review Clinic 4

Use this review clinic to strengthen a real unit from the book.

Practice context: small project or observation linked to Floating.

1. Re-read the lesson on Floating and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: observation is specific.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Glossary

Use this glossary to revise important words in Science and Technology.

- Classification of Plants: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- The Human Breathing System: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Matter 2.1 Mixtures Meaning of mixtures Types of mixtures (Winnowing, picking, sieving: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Floating: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Sound Energy: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Heat transfer: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.

Add five more words from your lessons, then write a meaning and one correct example sentence or worked example for each word.

Answer And Teacher Notes

A strong Science and Technology answer should:

- answer the exact question asked;
- separate observation from explanation;
- name materials, variables, safety checks, or data sources where needed;
- use labelled diagrams, tables, or step-by-step procedures when useful;
- draw a conclusion from evidence, not from guessing.

For investigations, good answers include a question, method, results, conclusion, and one limit or improvement. For technology tasks, good answers explain the design choice and how it solves the problem safely.

Answer-key guide: a complete science answer names the concept, shows evidence in a labelled sketch/table/model, explains the pattern, includes safety where relevant, and ends with a conclusion beginning from the evidence. Wrong answers often mix observation with opinion or leave out variables.

Rubric for self-checking:
- 4: The answer addresses the exact question, uses subject vocabulary correctly, gives evidence or working, and includes a useful check or correction.
- 3: The main answer is correct but one part needs stronger evidence, clearer working, better labels, or a fuller explanation.
- 2: Some understanding is shown, but the answer is incomplete, too general, or hard to follow.
- 1: The answer is guessed, off topic, or missing the evidence needed for the task.

Correction routine: reread the command word, underline the key data or evidence, improve one weak step, and write one sentence explaining why the final answer is reasonable.

Final Project

Create a final Science and Technology project.

Your project must include:

1. A clear question, problem, or design need.
2. Evidence from at least three lessons.
3. A safe method, labelled diagram, data table, flow chart, model, or prototype plan.
4. A conclusion that explains what the evidence shows.
5. One safety note, one limitation, and one improvement.

Present your project to a classmate, teacher, parent, or study group.